

Linux Administrator Interview Q&A — Prep Guide (0-2 Years Experience)

Based on the JD: Linux/Unix Administration, Bash/Shell Scripting, Linux Commands, Users & Groups, File Permissions, Process & Service Management, systemd, SSH, Networking, DNS, Disk & LVM Management, Package Management (YUM/Zypper), System Logs & Monitoring, Cron Jobs, Troubleshooting + Good-to-have: Kubernetes, Docker, RKE2/Rancher, Security & Hardening, Backup & Recovery, Cloud/Virtualization.

Each section is organized **Basic → Intermediate → Professional (2-yr level)**.

1. Linux/Unix Fundamentals

Q1. What is the difference between Linux and Unix? A: Unix is a proprietary OS family (AIX, Solaris, HP-UX) developed originally at Bell Labs. Linux is an open-source, Unix-like kernel created by Linus Torvalds, and distributions (RHEL, Ubuntu, SUSE) package it with GNU tools. Linux is free/open-source, highly customizable, and has broader community/enterprise support today.

Q2. What are the different runlevels/targets in Linux? A: Traditional SysV runlevels: 0 (halt), 1 (single-user), 2 (multi-user no NFS), 3 (multi-user with networking, no GUI), 5 (GUI), 6 (reboot). In systemd, these map to targets: poweroff.target, rescue.target, multi-user.target, graphical.target, reboot.target. Check current target with `systemctl get-default`.

Q3. What is the Linux boot process? A: BIOS/UEFI → Bootloader (GRUB2) → Kernel loads into memory → `initramfs` mounts temporary root → Kernel starts `systemd` (PID 1) → `systemd` reads targets and starts services → login prompt.

Q4. Explain the Linux directory structure (FHS). A: `/etc` (configs), `/bin`, `/sbin` (binaries), `/var` (logs, spool, variable data), `/home` (user dirs), `/tmp` (temp files), `/opt` (optional/third-party software), `/proc`, `/sys` (virtual, kernel/process info), `/boot` (kernel & bootloader files), `/mnt`, `/media` (mount points).

2. Linux Commands (Basic → Advanced)

Q5. Difference between `cp`, `mv`, and `rsync`? A: `cp` copies files; `mv` moves/renames; `rsync` synchronizes files/directories efficiently (only transfers differences), supports remote sync over SSH, and is preferred for backups (`rsync -avz source dest`).

Q6. How do you find files larger than 100MB in `/var/log`? A: `find /var/log -type f -size +100M`

Q7. How do you search for text inside multiple files? A: `grep -rn "pattern" /path/` — `-r` recursive, `-n` shows line numbers. For advanced filtering combine with `awk/sed`.

Q8. Difference between awk and sed? A: sed is a stream editor for find/replace and line-based edits (sed 's/old/new/g' file). awk is a pattern-scanning/field-processing language good for column-based data (awk '{print \$1,\$3}' file).

Q9. How do you check disk usage and inode usage? A: df -h (disk space), df -i (inode usage), du -sh /path/* (directory-level usage).

Q10 (Professional). A df -h shows 100% used but du -sh / shows much less space used — why, and how do you fix it? A: Usually caused by deleted files still held open by a running process (deleted-but-open file handles), or by hidden files/reserved blocks. Diagnose with lsof +L1 or lsof | grep deleted to find processes holding deleted file handles, then restart/kill the process to release space. Also check tune2fs -l for reserved root blocks (reserved block count).

3. Users, Groups & File Permissions

Q11. How do you create a user and add them to a secondary group? A: useradd -m -G developers chetan (creates home dir, adds to secondary group). Set password: passwd chetan. Modify later: usermod -aG groupname user (note -a = append, without it groups get overwritten).

Q12. Explain file permission types and how to read rwxr-xr--. A: r=read(4), w=write(2), x=execute(1). Owner=rwx(7), Group=r-x(5), Others=r-(4) → rwxr-xr-- = 754. Use chmod 754 file or chmod u+x,g-w file.

Q13. What is the difference between hard link and soft (symbolic) link? A: Hard link shares the same inode as the original file (can't cross filesystems, survives original deletion). Soft link is a pointer/path reference (can cross filesystems, breaks if original is deleted). Create: ln file hardlink, ln -s file softlink.

Q14. What are SUID, SGID, and Sticky bit? A: SUID (4xxx) — file runs with owner's privileges (e.g., /usr/bin/passwd). SGID (2xxx) — on directories, new files inherit the group. Sticky bit (1xxx) — on directories (e.g. /tmp), only the file owner can delete their own files. Set with chmod 4755, 2755, 1777 respectively.

Q15 (Professional). How do you audit and find all SUID files on a system, and why does it matter for security? A: find / -perm -4000 -type f 2>/dev/null. It matters because SUID binaries can be exploited for privilege escalation if misconfigured or vulnerable — part of routine hardening/audit checks.

Q16. What is umask and how does it work? A: umask defines default permission mask subtracted from max permissions (666 for files, 777 for dirs) when new files/dirs are created. Default umask 022 → new files get 644, new dirs get 755. Check/set with umask / umask 027.

4. Process & Service Management

Q17. Difference between a process and a thread? A: A process is an independent execution unit with its own memory space; a thread is a lightweight execution unit within a process sharing memory with other threads of the same process.

Q18. How do you check running processes and kill one? A: `ps aux`, `top`/`htop` for live view; find PID and `kill -9 <PID>` (force kill) or `kill -15 <PID>` (graceful, `SIGTERM`). Prefer `SIGTERM` before `SIGKILL`.

Q19. Difference between `kill -9` and `kill -15`? A: `-15` (`SIGTERM`) asks the process to terminate gracefully, allowing cleanup. `-9` (`SIGKILL`) forcibly terminates immediately without cleanup — used only when a process is unresponsive.

Q20. What is a zombie process and how do you deal with it? A: A zombie is a terminated process whose exit status hasn't been read by its parent, so its entry lingers in the process table. Identify with `ps aux | grep Z`. Fix: signal or restart the parent process (killing the zombie itself does nothing since it's already dead).

Q21 (Professional). Server load average is high but CPU usage looks normal — what do you check? A: Load average includes processes waiting on I/O, not just CPU. Check `iostat -x 1`, `vmstat 1`, `top` (look at `wa` — I/O wait %), and `iotop` to find processes stuck on disk I/O. Could indicate slow disk, storage saturation, or NFS latency.

5. systemd & Service Management

Q22. How do you start, stop, enable, and check status of a service using systemd? A: `systemctl start httpd`, `systemctl stop httpd`, `systemctl enable httpd` (start at boot), `systemctl status httpd`, `systemctl restart httpd`, `systemctl daemon-reload` (after unit file changes).

Q23. What is a systemd unit file and what are its main sections? A: A config file (`.service`, `.timer`, `.mount`, etc.) usually in `/etc/systemd/system/`. Main sections: `[Unit]` (description, dependencies like `After=`, `Requires=`), `[Service]` (`ExecStart`, `Type`, `Restart` policy), `[Install]` (`WantedBy` target).

Q24. How do you create a custom systemd service for an application? A: Create `/etc/systemd/system/myapp.service` with `ExecStart` pointing to the binary/script, run `systemctl daemon-reload`, then `systemctl enable --now myapp`.

Q25 (Professional). A service keeps crash-looping — how do you troubleshoot with systemd? A: `systemctl status myapp` for last error/exit code, `journalctl -u myapp -xe` for full logs, check `Restart=` and `RestartSec=` policy in the unit file to avoid rapid restart storms, verify dependencies (`After=`, `Requires=`) and permissions of `ExecStart` binary.

6. SSH & Remote Access

Q26. How does SSH key-based authentication work? A: A key pair is generated (`ssh-keygen`) — private key stays on client, public key is copied to the server's `~/.ssh/authorized_keys` (`ssh-copy-id user@host`). The server encrypts a challenge with the public key; only the matching private key can decrypt it, authenticating the user without a password.

Q27. How do you harden SSH access? A: Disable root login (`PermitRootLogin no`), disable password auth (`PasswordAuthentication no`, use keys only), change default port, use `AllowUsers/AllowGroups`, enable `fail2ban`, restrict via firewall, use `Protocol 2` only, and keep `sshd` updated.

Q28. How do you copy files securely between servers? A: `scp` file `user@host:/path/`, or `rsync -avz -e ssh source user@host:/dest/` for large/incremental transfers.

Q29 (Professional). SSH connection hangs/times out intermittently — how do you troubleshoot? A: Check network/firewall (`telnet host 22` or `nc -zv host 22`), check `TCPKeepAlive/ClientAliveInterval` settings in `sshd_config`, check server load/resource exhaustion, review `/var/log/secure` or `journalctl -u sshd`, and check DNS reverse-lookup delays (`UseDNS no` can help).

7. Networking

Q30. How do you check IP configuration and connectivity in Linux? A: `ip a` / `ip addr show` (interfaces), `ip route` (routing table), `ping`, `traceroute`/`tracert`, `netstat -tulnp` or `ss -tulnp` (listening ports), `curl -I url` (HTTP check).

Q31. Difference between TCP and UDP? A: TCP is connection-oriented, reliable, ordered (handshake, retransmission) — used for HTTP, SSH, FTP. UDP is connectionless, faster, no guaranteed delivery — used for DNS queries, streaming, VoIP.

Q32. What is the difference between netstat and ss? A: `ss` is the modern replacement for `netstat`, faster because it reads directly from kernel space (via `netlink`) instead of parsing `/proc`. Both show socket/connection info, but `ss` is preferred on current systems.

Q33. How do you troubleshoot “cannot reach a remote server” issue step by step? A: 1) ping the host (check basic reachability/ICMP). 2) `traceroute` to find where it breaks. 3) Check DNS resolution `nslookup/dig`. 4) Check specific port with `nc -zv host port` or `telnet`. 5) Check local firewall (`iptables/firewalld`) and remote firewall/security groups. 6) Check `/etc/hosts` and routing table.

Q34 (Professional). How would you configure a static IP and persistent network settings on RHEL/CentOS vs Ubuntu? A: RHEL/CentOS: edit `/etc/sysconfig/network-scripts/ifcfg-<iface>` (or `nmcli con mod` with `NetworkManager`) then `nmcli con up <iface>`. Ubuntu (netplan): edit `/etc/netplan/*.yaml` with static IP/gateway/DNS, then `netplan apply`.

8. DNS

Q35. How does DNS resolution work? A: Client queries local resolver → recursive DNS server → root servers → TLD servers → authoritative name server returns the IP, which gets cached along the chain based on TTL.

Q36. Difference between A, CNAME, MX, and PTR records? A: A = maps hostname to IPv4. CNAME = alias pointing to another hostname. MX = mail server priority for a domain. PTR = reverse lookup, maps IP back to hostname.

Q37. How do you test/troubleshoot DNS resolution issues on a Linux server? A: `dig domain.com`, `nslookup domain.com`, check `/etc/resolv.conf` for correct nameservers, check `/etc/nsswitch.conf` ordering, check `/etc/hosts` for conflicting static entries, and test with `dig @8.8.8.8 domain.com` to isolate whether it's the resolver or the DNS server.

9. Disk & LVM Management

Q38. What is LVM and why is it used? A: Logical Volume Manager abstracts physical storage into Physical Volumes (PV) → Volume Groups (VG) → Logical Volumes (LV), allowing flexible resizing, spanning multiple disks, and easier storage management than raw partitions.

Q39. Walk through creating an LVM volume from scratch. A:

```
pvccreate /dev/sdb
vgcreate vg_data /dev/sdb
lvcreate -L 10G -n lv_app vg_data
mkfs.xfs /dev/vg_data/lv_app
mount /dev/vg_data/lv_app /app
```

Add to /etc/fstab for persistence.

Q40. How do you extend a logical volume and filesystem without downtime? A:

```
lvextend -L +5G /dev/vg_data/lv_app
xfs_growfs /app          # for XFS
# or resize2fs /dev/vg_data/lv_app for ext4
```

Q41. Difference between mkfs.xfs and mkfs.ext4? A: XFS is default on RHEL 7+, better for large files/high performance, but cannot be shrunk. ext4 is more universal, supports shrinking, mature and widely compatible.

Q42 (Professional). A disk shows “No space left on device” but df -h shows free space — what’s happening and how do you fix it? A: Likely inode exhaustion. Check df -i. If inodes are at 100%, too many small files exist (e.g., session/cache files). Fix by clearing unnecessary small files; for ext-based filesystems inode count is fixed at creation time (would need reformat to change), so cleanup or migrating to XFS (dynamic inodes) helps long term.

Q43. How do you check and repair a filesystem? A: Unmount first, then fsck /dev/sdX1 (ext) or xfs_repair /dev/sdX1 (XFS). Never run fsck on a mounted filesystem.

10. Package Management

Q44. Difference between YUM/DNF and Zypper? A: YUM/DNF are used on RHEL/CentOS/Fedora (RPM-based); Zypper is used on SUSE/openSUSE (also RPM-based, but with different dependency resolver and repo tooling). Both handle install/update/remove/dependency resolution for RPM packages.

Q45. Common package management commands (YUM/DNF)? A: yum install pkg, yum remove pkg, yum update, yum list installed, yum search keyword, yum info pkg, yum repolist, rpm -qa | grep pkg (query installed RPMs directly).

Q46. Common Zypper commands? A: zypper install pkg, zypper remove pkg, zypper update, zypper refresh (update repo metadata), zypper search keyword, zypper info pkg.

Q47 (Professional). How do you set up and manage a local/offline YUM repository? A: Mount the ISO or copy RPMs to a directory, run createrepo /path/to/repo, create a .repo file in /etc/yum.repos.d/ pointing baseurl=file:///path/to/repo, set gpgcheck as needed, then yum clean all && yum repolist.

11. System Logs & Monitoring

Q48. Where are system logs stored and how do you view them?

A: Traditional syslog: /var/log/messages, /var/log/secure, /var/log/cron. On systemd systems: journalctl (binary journal) — journalctl -xe (recent errors), journalctl -u servicename, journalctl -since "1 hour ago", journalctl -f (live tail, like tail -f).

Q49. How do you monitor real-time system resource usage?

A: top/htop (CPU/mem), vmstat (memory/CPU/IO summary), iostat (disk I/O), sar (historical performance data via sysstat), free -h (memory), netstat/ss (network).

Q50. What is log rotation and how is it configured? A: logrotate manages log file size/retention by rotating, compressing, and deleting old logs. Configured in /etc/logrotate.conf and /etc/logrotate.d/*, run via cron/systemd timer. Key directives: rotate, daily/weekly, compress, size, maxage.

Q51 (Professional). A server ran out of disk space overnight — how do you find the root cause after the fact? A: Check log growth via journalctl --disk-usage, review /var/log file sizes (du -sh /var/log/*), check for logrotate failures, review cron job outputs, check application logs for error loops that could be flooding logs, and consider setting up monitoring/alerting (disk threshold alerts) to catch it earlier next time.

12. Cron Jobs & Automation

Q52. Explain crontab syntax. A: * * * * * command → minute(0-59) hour(0-23) day-of-month(1-31) month(1-12) day-of-week(0-7). E.g., 0 2 * * * /scripts/backup.sh runs daily at 2 AM.

Q53. Difference between user crontab and /etc/cron.d? A: User crontab (crontab -e) is per-user, doesn't specify user in the line. System-wide cron files in /etc/cron.d/ or /etc/crontab require specifying the user to run as, and are managed via config files (good for packaged/automated deployment).

Q54. How do you debug a cron job that isn't running as expected? A: Check /var/log/cron for execution logs, ensure the script has execute permission, use absolute paths in the script (cron has a minimal PATH env), redirect output to a log file for debugging (>> /var/log/myjob.log 2>&1), and confirm the cron daemon (crond/cron) is running.

13. Bash/Shell Scripting

Q55. Write a script to check disk usage and alert if above 80%.

A:

```
#!/bin/bash
THRESHOLD=80
USAGE=$(df -h / | awk 'NR==2 {print $5}' | tr -d '%')
if [ "$USAGE" -gt "$THRESHOLD" ]; then
    echo "Warning: Disk usage is at ${USAGE}%" | mail -s "Disk Alert"
admin@example.com
fi
```

Q56. Difference between \$* and \$@ in shell scripting? A: Both represent all positional parameters, but "\$@" expands each argument as a separate quoted word (preserves spaces correctly), while "\$*" expands them as a single string. "\$@" is preferred in loops.

Q57. How do you take user input and use conditionals/loops in bash? A:

```
#!/bin/bash
read -p "Enter service name: " svc
for i in 1 2 3; do
    if systemctl is-active --quiet "$svc"; then
        echo "$svc is running"
        break
    else
        systemctl restart "$svc"
        sleep 5
    fi
done
```

Q58 (Professional). Write a script to find and archive log files older than 30 days. A:

```
#!/bin/bash
LOGDIR="/var/log/myapp"
find "$LOGDIR" -type f -name "*.log" -mtime +30 -exec gzip {} \;
find "$LOGDIR" -type f -name "*.log.gz" -mtime +90 -delete
```

14. Troubleshooting Scenarios (Common Interview Style)

Q59. Server is running slow — walk through your troubleshooting approach. A: 1) top/htop — check CPU, memory, load average. 2) free -h — memory/swap usage. 3) iostat/iotop — disk I/O bottleneck. 4) df -h — disk space. 5) netstat/ss — network connections/saturation. 6) Check logs (journalctl, /var/log/messages) for errors. 7) Identify top resource-consuming process and act (restart/kill/scale).

Q60. A user can't log in via SSH — how do you troubleshoot? A: Check if the service is up (systemctl status sshd), check firewall/port 22 open, check /var/log/secure or journalctl -u sshd for auth failures, verify account isn't locked (passwd -S user), check sshd_config restrictions (AllowUsers, PermitRootLogin), verify correct key/permissions on ~/.ssh (should be 700 for dir, 600 for authorized_keys).

Q61. How do you recover a system if /etc/fstab is misconfigured and the server won't boot? A: Boot into rescue/single-user mode (or GRUB kernel parameter init=/bin/bash / systemd.unit=rescue.target), remount root as read-write (mount -o remount,rw /), edit /etc/fstab to fix the faulty entry, then reboot.

15. Good-to-Have: Kubernetes, Docker & RKE2/Rancher

Q62. What is the difference between a Docker container and a VM? A: Containers share the host OS kernel and virtualize at the process level (lightweight, fast startup), while VMs virtualize hardware and run a full separate OS (heavier, more isolated). Docker uses namespaces and cgroups for isolation.

Q63. Basic Docker commands you should know? A: `docker ps`, `docker images`, `docker run -d -p 8080:80 nginx`, `docker exec -it <container> bash`, `docker logs <container>`, `docker build -t myimage .`, `docker-compose up -d`.

Q64. What is Kubernetes and its core components? A: An orchestration platform for managing containerized applications at scale. Core components: **Control Plane** (API server, etcd, scheduler, controller-manager) and **Worker Nodes** (kubelet, kube-proxy, container runtime). Key objects: Pod, Deployment, Service, ConfigMap, Secret, Namespace.

Q65. What is RKE2/Rancher and why would an org use it? A: RKE2 (Rancher Kubernetes Engine 2) is a lightweight, security-hardened Kubernetes distribution by SUSE/Rancher, CIS-benchmark compliant by default — often used in regulated/enterprise environments. Rancher is the management UI/control plane for managing multiple Kubernetes clusters (RKE2, EKS, AKS, GKE) from a single pane of glass.

Q66. Basic kubectl commands? A: `kubectl get pods`, `kubectl describe pod <name>`, `kubectl logs <pod>`, `kubectl apply -f file.yaml`, `kubectl exec -it <pod> -- bash`, `kubectl get nodes`, `kubectl rollout restart deployment/<name>`.

16. Good-to-Have: Security, Hardening, Backup & Cloud

Q67. Name basic Linux hardening steps you'd apply on a new server. A: Disable root SSH login, enforce key-based SSH auth, disable unused services, configure firewall (firewalld/iptables), apply regular patching, enforce password policies (chage, PAM), set up SELinux/AppArmor in enforcing mode, remove unnecessary SUID binaries, enable auditd for logging, restrict sudo access with least privilege.

Q68. What is SELinux and what are its modes? A: SELinux (Security-Enhanced Linux) enforces Mandatory Access Control policies beyond standard permissions. Modes: **Enforcing** (blocks and logs violations), **Permissive** (logs only, doesn't block — useful for troubleshooting), **Disabled**. Check with `getenforce`, set with `setenforce 0/1`, permanent change in `/etc/selinux/config`.

Q69. Difference between full, incremental, and differential backups? A: Full = complete copy every time (most storage, simplest restore). Incremental = only changes since the last backup of any type (least storage, but restore requires the full chain). Differential = changes since the last full backup (middle ground — faster restore than incremental, more storage than incremental).

Q70. How would you approach setting up automated backups for a Linux server? A: Use `rsync`/`tar` scripts scheduled via `cron`, or tools like Bacula/Restic/Veeam for enterprise setups; store backups offsite/in cloud (S3, etc.); test restores periodically; follow 3-2-1 rule (3 copies, 2 media types, 1 offsite).

Q71. Basic cloud/virtualization concepts you should be familiar with? A: Virtualization: hypervisors (KVM, VMware ESXi, Hyper-V), VM provisioning, snapshots. Cloud (AWS as example): EC2 (compute), S3 (storage), IAM (access), VPC (networking), Security Groups (firewall), basic knowledge of provisioning and scaling compute resources.

Quick Revision Checklist (Day Before Interview)

- ☐ Practice chmod/chown/umask numeric + symbolic notation
- ☐ Be ready to write a basic bash script live (loop + conditional + cron)
- ☐ Know LVM commands (pv/vg/lv create, extend) by heart
- ☐ Be able to explain a real troubleshooting incident from your Atos experience (STAR format)
- ☐ Review systemctl/journalctl commands
- ☐ Review basic kubectl/docker commands since you have hands-on AWS/Docker/K8s exposure
- ☐ Prepare 2-3 questions to ask the interviewer about their environment (RHEL vs SUSE, on-prem vs cloud, RKE2/Rancher usage)

Good luck with your interview, Chetan!